

Random Edge Sampling for Spanner Construction

B.Sc. Thesis in Computer Science

Candidate: Mohammadreza Ahmadi

Supervisor: Dr. Hossein Jowhari

Faculty of Mathematics
K. N. Toosi University of Technology

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Core Definitions: Graph Spanners

We consider unweighted, undirected n -vertex graphs $G = (V, E)$.

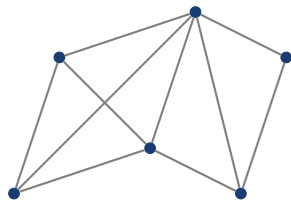
- **Multiplicative α -Spanner:** A subgraph $H = (V, E')$ preserving shortest paths up to a **stretch factor** $\alpha \geq 1$:

$$d_H(u, v) \leq \alpha \cdot d_G(u, v) \quad \forall u, v \in V$$

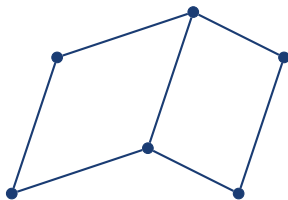
Proposition 1

H is a t -spanner of $G \iff d_H(u, v) \leq t$ for all edges $uv \in E$.

Example: A 2-Spanner



Original Graph G



A 2-Spanner H

- **Intractability:** Finding the optimally sparse t -spanner is **NP-Hard**, forcing us to use approximations.

Theoretical Limits: Girth and Edge Density

Girth (g): Length of the shortest cycle in a graph.

Proposition 2

*A graph with girth $> t + 1$ has **no proper subgraph** that is a t -spanner.*

Maximum Edge Density (γ):

- $\gamma(n, k)$: Max possible edges in an n -vertex graph with girth $> k$.
- **The Fundamental Barrier:** Because of the proposition above, no algorithm can construct a t -spanner sparser than $\gamma(n, t + 1)$ in the worst case.

The Asymptotic Baseline: Moore & Erdős

Exact values for $\gamma(n, k)$ remain a major open problem, but we rely on asymptotic bounds:

Moore Bound & Erdős' Girth Conjecture

For graphs with girth $> 2k$:

- **Upper Bound (Moore):** $O(n^{1+1/k})$ edges.
- **Lower Bound (Erdős):** $\Omega(n^{1+1/k})$ edges (Conjectured tight).
- A $(2k - 1)$ -spanner with $O(n^{1+1/k})$ edges is considered structurally **optimal**.

The Classical Approach: Greedy Algorithm

Greedy Spanner (Althöfer et al., 1993)

Process edges sequentially. Add $e = (u, v)$ to H **only if**:

$$d_H(u, v) > 2k - 1$$

Pros

✓ Optimal Sparsity

Guarantees $O(n^{1+1/k})$ edges.

Cons

✗ High Time Complexity

$O(k \cdot n^{2+1/k})$ bottleneck.

✗ Strictly Sequential

Fails in distributed networks.

State of the Art: Elkin & Neiman (2019)

Goal: Bypass the Greedy algorithm's sequential bottleneck via a distributed randomized framework.

The Mechanism: **Vertex-Based Randomness**

- Each vertex $u \in V$ independently samples $r_u \sim \mathcal{EX}\mathcal{P}(\ln(cn)/k)$.
- Vertices broadcast these radii to neighbors and select edges based on the maximums received.

Theoretical Achievements:

- Achieves $O((n/\epsilon)^{1+1/k})$ expected edges.
- Computes the spanner in exactly k communication rounds (CONGEST model).

The Paradigm Shift: Our Motivation

The Elkin-Neiman framework is elegant, but inherently algorithmic:

- Randomness is exclusively assigned to the **vertices**.
- It requires message passing, broadcasting, and state comparisons.

The Core Question of Our Thesis

Can we eliminate these communication phases entirely?

*What if we sample edges **independently and directly**, relying solely on the local topology of the graph?*

Independent Edge Sampling: Initial Attempts

How do we define the independent sampling probability p_e ?

Attempt 1: Global Minimum Probability

- Set a uniform, high probability across all edges to ensure path survival.
- **Result:** The graph remains far too dense (fails the sparsity goal).

Attempt 2: Deterministic Deletion

- Drop an edge e simply because it has an alternative path P .
- **Result:** Edges inside P drop independently (cascading failures). The path collapses, destroying the stretch guarantee.

The Dependency Bottleneck

Definition: Let x_e be the maximum number of edge-disjoint **good paths** (length $\leq t$) alternative to e .

$$p_e = \begin{cases} 1 & \text{if } x_e \leq c \\ \frac{c}{x_e} & \text{if } x_e > c \end{cases}$$

Result:

- Path survival requires *every* internal edge to be sampled.
- A single dense edge ($x_{e'} \gg c$) in a good path has $p_{e'} \rightarrow 0$.
- This single dropped edge disconnects the path and violates the stretch.

Our Contribution: The $2c$ -Cutoff Filter

Core Idea: Alternative paths must be structurally **reliable**.

Definition: Safe Path

A path P is a **safe path** for edge e if:

- ① Edge-disjoint from all other safe paths for e .
- ② Length $\leq t = 2k - 1$.
- ③ Contains no highly dense edges: $x_{e'} \leq 2c$ ($\forall e' \in P$).

To ensure path survival, the cutoff is set to:

$$c = \Theta(2^t \log n)$$

Expected Sparsity Analysis

Goal: Bound the expected number of edges $\mathbb{E}[|E'|] = \sum p_e$.

1. Critical Edges ($x_e \leq c$) $\implies p_e = 1$

- Bounded via Greedy Decomposition and Moore's Bound.
- **Total size:** $O(c \cdot n^{1+1/k})$.

2. Dense Edges ($x_e > c$) $\implies p_e = c/x_e$

- Partitioned into $O(\log n)$ exponentially growing buckets.
- Bucket growth rate perfectly cancels the exponential drop in p_e .
- **Total size:** $O(c \cdot n^{1+1/k} \log n)$.

Final Spanner Size & Performance

Substituting $c = \Theta(2^{2k} \log n)$, the expected number of edges is:

$$O(2^{2k} n^{1+1/k} \log^2 n)$$

Constant Stretch Regime:

- For small constant k , the 2^{2k} overhead is a trivial factor.
- The framework naturally produces highly sparse spanners purely through independent sampling.

Optimal Scaling:

- By optimizing the scale parameter to $k^* = \Theta(\sqrt{\log n})$.
- The total expected size of the spanner becomes near-linear: $n^{1+o(1)}$.

The Algorithmic Challenge

The NP-Hardness

Computing x_e (the exact maximum number of edge-disjoint safe paths) is **NP-Hard** for $t \geq 4$ (Itai et al., 1982).

We need an efficiently computable parameter q_e that can safely replace x_e without violating the stretch guarantees.

The Two-Phase Greedy Algorithm

Phase 1: Upper Bound Extraction

- Temporarily **ignore** the safety condition.
- Run iterative BFS to extract a maximal set of paths. Let this count be $q_e^{(0)}$.
- If $q_e^{(0)} \leq 2c$, the edge is definitively labeled **safe**. Otherwise, it is **heavy**.

Phase 2: Safe Path Isolation

- Remove all *heavy* edges from the graph.
- Re-run iterative BFS on this purified subgraph.
- The new count is q_e . Every path found is structurally guaranteed to be valid and safe.

Time Complexity: Iterative BFS takes $O(m)$ per path $\Rightarrow \mathbf{O(m^3)}$ overall.

Algorithmic Realization: Greedy Approximation

Lemma: Greedy t -Approximation

The extracted maximal set size (q_e) bounds the optimal maximum set size (x_e):

$$\frac{x_e}{t} \leq q_e \leq x_e$$

Proof:

- Let OPT ($|OPT| = x_e$) be the *maximum* possible packing of safe paths.
- Let $GREEDY$ ($|GREEDY| = q_e$) be our extracted *maximal* packing.

The Upper Bound ($q_e \leq x_e$)

- Trivial by definition. Since x_e is the absolute theoretical maximum capacity, no valid packing ($GREEDY$) can physically exceed it.

Proof Continued: The Lower Bound ($x_e \leq t \cdot q_e$)

The Lower Bound

- Since *GREEDY* is *maximal*, it cannot be extended. Thus, every path in *OPT* must intersect at least one path in *GREEDY*.
- A single path $P \in \text{GREEDY}$ has at most t edges.
- Because paths in *OPT* are strictly edge-disjoint, each edge of P can intersect at most *one* path in *OPT*.
- Consequently, a single path in *GREEDY* can block no more than t distinct paths in *OPT*.

Conclusion

Summing the maximum blocked paths across all of *GREEDY*:

$$|OPT| \leq t \cdot |GREEDY| \implies x_e \leq t \cdot q_e \square$$

The Final Algorithmic Spanner Size

Since $q_e \geq x_e/t$, the approximation increases the sampling probability by at most a factor of t .

Linearity of Expectation:

$$\mathbb{E}[|E'|] = \sum_{e \in E} p'_e \leq t \sum_{e \in E} p_e = t \cdot \mathbb{E}[|E_{opt}|]$$

Polynomial-Time Guarantee

Substituting $t = 2k - 1$, the final expected spanner size becomes:

$$O(k \cdot 2^{2k} n^{1+1/k} \log^2 n)$$

Conclusion & Open Problems

Key Contributions:

- Designed a novel framework based purely on independent edge sampling.
- Resolved the dependency bottleneck using the $2c$ -cutoff filter.
- Bypassed NP-Hardness via an $O(m^3)$ two-phase greedy approximation.
- Achieved an expected spanner size of $O(k \cdot 2^{2k} n^{1+1/k} \log^2 n)$.

Future Directions:

- Is the $O(2^{2k})$ overhead fundamental to independent sampling, or merely an artifact of our analysis?
- Can the $O(m^3)$ time complexity or the t -approximation ratio be further improved?
- Are there entirely different local topological metrics that avoid path-counting (x_e) altogether?

Thank You!